

Experiment 1

Aim

To understand the working of the following: (i) IP Address, (ii) Cisco IOS, (iii) Straight-Through Cable, Cross-Over Cable, and RJ45 Connector, (iv) Layer-2 Switch, and (v) Router.

Objective

1. To understand the concept and importance of IP addressing in computer networks.
2. To study the architecture and command-line interface of Cisco IOS along with its operating modes.
3. To differentiate between Straight-Through and Cross-Over Ethernet cables and identify their practical applications.
4. To understand the structure and purpose of the RJ45 connector used in Ethernet communication.
5. To study the working principle of a Layer-2 Switch and its role in forwarding frames based on MAC addresses.
6. To understand the functions of a Router and its role in interconnecting different networks using IP addresses.
7. To gain hands-on experience with Cisco Packet Tracer for network design, configuration, and simulation.
8. To verify network connectivity using basic troubleshooting tools such as the ping command.

Practical / Experiment Steps

1. Open Cisco Packet Tracer on the computer system.
2. Study the concepts of IPv4, IPv6, subnet mask, network ID, and host ID.
3. Create a simple network topology by connecting two PCs and assign valid IP addresses.
4. Test communication between the PCs using the ping command.
5. Add a Cisco Router and access the Cisco IOS Command-Line Interface (CLI).
6. Explore the different Cisco IOS modes, including User EXEC Mode, Privileged EXEC Mode, and Global Configuration Mode.
7. Execute basic Cisco IOS commands such as **show version**, **show running-config**, **show ip interface brief**, **ping**, and **configure terminal**.
8. Study the construction and applications of Straight-Through Cable, Cross-Over Cable, and the RJ45 Connector.
9. Create a network topology using a Layer-2 Switch and observe frame forwarding based on MAC addresses.
10. Observe the simulation results, verify successful communication between network devices, and save the Packet Tracer project.

Procedure of the Experiment

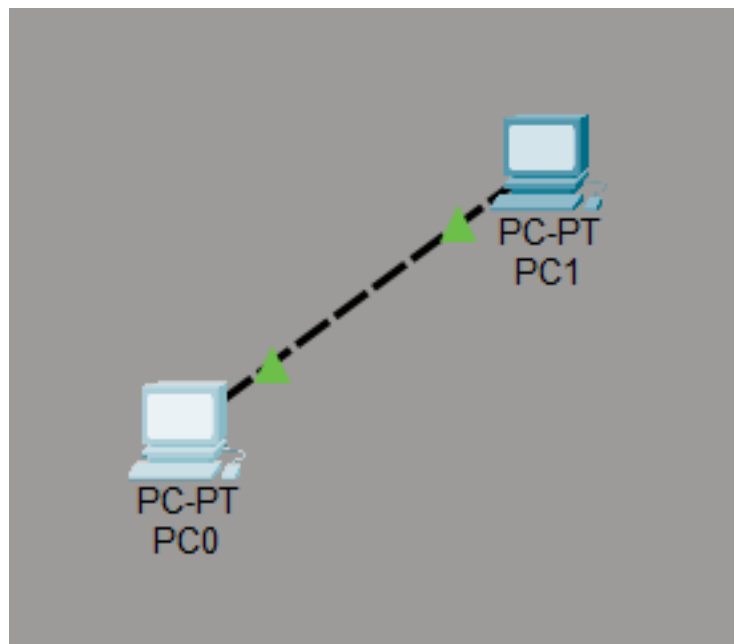
1. Start the computer system and launch Cisco Packet Tracer.

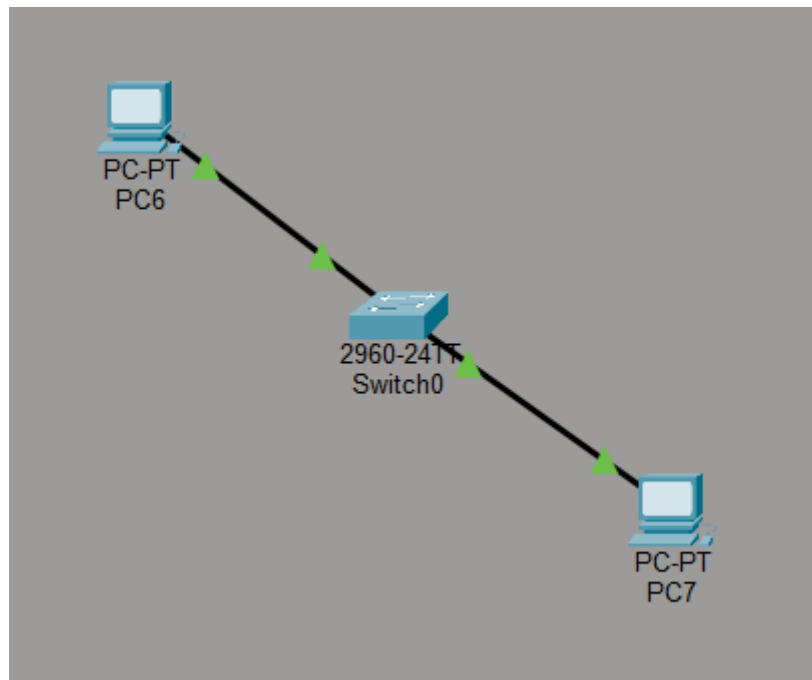
2. Create a new workspace for the experiment.
3. Place two PCs in the workspace and connect them using the appropriate networking device (direct connection or a Layer-2 Switch).
4. Assign valid IP addresses and subnet masks to both PCs through the **Desktop → IP Configuration** option.
5. Verify network connectivity by opening the Command Prompt and executing the **ping** command.
6. Add a Cisco Router to the topology and access its Command-Line Interface (CLI).
7. Explore the different Cisco IOS modes, including User EXEC Mode, Privileged EXEC Mode, and Global Configuration Mode, and execute basic IOS commands.
8. Study the structure and applications of Straight-Through Cable, Cross-Over Cable, and the RJ45 Connector used in Ethernet networking.
9. Create a network topology using a Layer-2 Switch and observe how it forwards frames based on MAC addresses.
10. Analyze the simulation, verify successful communication between devices, record the observations, and save the Packet Tracer project along with the required screenshots.

Input / Output Details

Input

Network topology, IP addresses, Ethernet cables, and Cisco IOS commands configured in Cisco Packet Tracer.





Output

Successful communication between network devices and verification of IP addressing, Cisco IOS commands, switching, and routing concepts through Cisco Packet Tracer.

Learning Outcome

After completing this experiment, the student was able to understand the concepts of IP addressing, Cisco IOS, Ethernet cables, RJ45 connectors, Layer-2 switches, and routers. The experiment also provided practical experience in configuring basic network devices, verifying network connectivity, and analyzing communication using Cisco Packet Tracer.